Random forest modelling as a tool for propagation and bend excess loss minimization on silicon nitride waveguide platforms

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ABSTRACT

This paper presents a detailed investigation into the propagation loss characteristics of silicon nitride strip waveguides at an 850 nm wavelength, utilizing a random forest model. The primary aim is to optimize low-loss conditions in photonic integrated circuits (PICs). To achieve this, a systematic 2x3 full factorial design of experiments is implemented, focusing on different layers within the PIC framework. The study revolves around a critical examination of how the waveguide width influences propagation loss. Leveraging the random forest model, known for its high precision in complex data analysis, we delve into the correlation between various design elements and their impact on loss. This methodology not only aids in pinpointing the pivotal factors affecting loss but also elucidates their interplay, particularly emphasizing the role of waveguide width. One of the key contributions of this research is the identification of optimal material configurations that significantly reduce loss. This is instrumental in enhancing the efficiency of PICs, a crucial aspect for their performance in applications such as optical communications and photonic computing. Our approach uniquely combines empirical data analysis with machine learning techniques, offering a novel perspective in photonic engineering research. The findings of this study not only shed light on the complex dynamics of waveguide design but also pave the way for the development of more efficient and effective photonic systems. This research stands to make a significant impact in the field, presenting a comprehensive methodology for designing low-loss silicon nitride strip waveguides, thereby contributing to the advancement of photonic technologies.

Keywords: Silicon Nitride Waveguides, Random Forest Modelling, Design of experiment,Loss optimization, NIR , Sensing

1. INTRODUCTION

The field of integrated photonics, which concentrates on the control and transmission of light at the micro and nano scales, has witnessed a remarkable growth in recent times. Integral to advancements in high-speed telecommunications, progressive computing technologies, and complex sensing mechanisms, integrated photonics stands as a pivotal contributor to technological innovations. Photonic Integrated Circuits (PICs), at the forefront of these advancements, have transformed the landscape of high-speed detection in biosensing applications and beyond. Nevertheless, a major challenge remains: the reduction of optical losses in mass-produced devices, primarily due to losses caused by surface roughness-induced scattering and absorption.[1–3]

This investigation explores Si_3N_4 waveguides, celebrated for their compatibility with both high-temperature front-end and low-temperature back-end processing techniques. The study assesses these waveguides over a variety of widths and bend radii in the infrared spectrum. Through comprehensive analysis and an automated setup for wafer-level measurements, the research aims to unravel the complexities of how various Si_3N_4 waveguide and cladding layer combinations interact, with applications in sensing and spectroscopy in mind.[4,5]

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Central to our analysis is the Design of Experiments (DOE), which is instrumental in examining the effects and interactions of multiple variables in the wafer-level evaluation of integrated structures, particularly concerning propagation and bend excess losses. Employing a full-factorial design to explore every factor combination, the study methodically uncovers the primary and interaction effects on losses, facilitating the identification of optimal configurations for PICs to certain peformance losses, crucial for mass production in the integrated photonics realm. In a previous study classification and regression trees (CART) are utilized to approach this task.[6]

To navigate the complexities of PIC design more effectively, a profound knowledge of photonics is married with the application of advanced statistical techniques. This study adopts a comprehensive approach that blends traditional photonics research with the sophisticated analytical power of statistical learning, epitomized by Random Forests, an ensemble method that builds upon CART by using multiple decision trees to improve predictive accuracy and control over-fitting, further enhances this analysis. It allows for a more robust and nuanced interpretation of how layer characteristics such as roughness, absorption, thickness, and the refractive index influence the propagation and bend excess losses. This detailed regression analysis is vital for understanding how individual layers and their interactions affect the overall performance of PICs.[7]

This paper aims not only to investigate the dependence of optical losses on layer characteristics but also to introduce a predictive maintenance methodology that Random Forests with the DOE to evaluate the predictability of the performance of such models. This approach offers an in-depth understanding of the roles of different layers and their complex interplays, advancing a flexible prediction the performance of loss-related properties like the sensitivity of waveguide-based sensors.

2. DESIGN, STRUCTURE AND MEASUREMENT OF TEST CONFIGURATIONS

Designing suitable test configurations to employ losses as response variables is imperative. The platform's intended wavelength for sensing applications is 850 nm, with a preference for transverse-electric (TE) polarization. Waveguide losses arise from factors such as material absorption and scattering, notably interface scattering due to surface roughness in standard passive entities like strip waveguides.[1,6]

In preparation for potential future applications in chemo/biosensing, an investigation was undertaken, examining waveguide widths spanning from 350 to 750 nm. Operating within the single-mode regime at the specified height and wavelength, these widths exhibit minimal propagation losses. Additionally, they enhance the field maximum around the waveguide core during TE polarization, improving the precision of biosensing measurements.[6,8]

Employing Synopsis Optodesigner, unique test configurations were carefully devised. The evaluation of propagation loss involved the application of the cutback method, employing waveguide spirals with different lengths, all maintaining a constant bend radius of 50 μm. Bend excess loss was examined using spirals of different radii, ranging from 10 to 35 μm, and a structure with a radius of 70 μm for comparison. All structures maintained a consistent path length of approximately 0.92 cm and widths varying from 450 to 750 nm.[6]

Testing integrated photonic structures at the wafer level demands precision, especially with components like single-mode fibres (SMFs) requiring meticulous alignment. Employing the FormFactor Cascade SUMMIT200 probe system, designed for 200 mm wafers, ensured (semi-)automated testing. The system, equipped with a silicon photonics module, facilitated accurate SMF placement from OZ Optics (SMF-633-4/125), ensuring consistent performance throughout testing. Grating couplers (GC) with a 10° coupling angle served as the primary coupling mechanism, optimized using an advanced piezoelectric stage for ultra-fine adjustments. The measurement setup included a near-infrared diode laser 852 nm, a polarization controller (Thorlabs FPC030), and a versatile power meter (Newport 2101). FormFactor Velox 3.0 software, FormFactor SiPh-Tool, and FormFactor FFI Photonics interface facilitated cohesive operations.[6] as depicted in Figure 1. Measurements corresponding to layouts focused on dies marked in red as delineated in Figure 2.

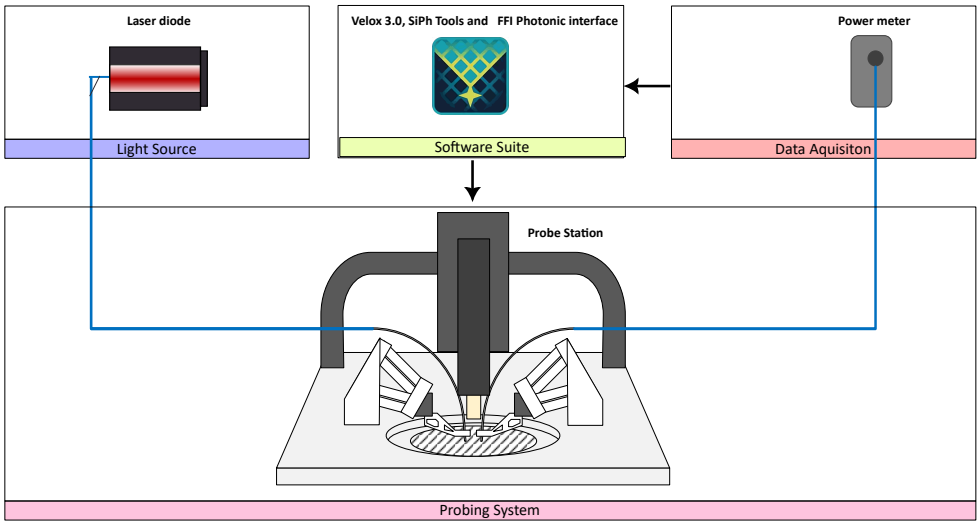


Figure 1. Conceptual representation of the experimental arrangement for conducting wafer-level assessments of propagation loss and bend loss configurations, encompassing a probe station, light source, power meter, and a software control system.

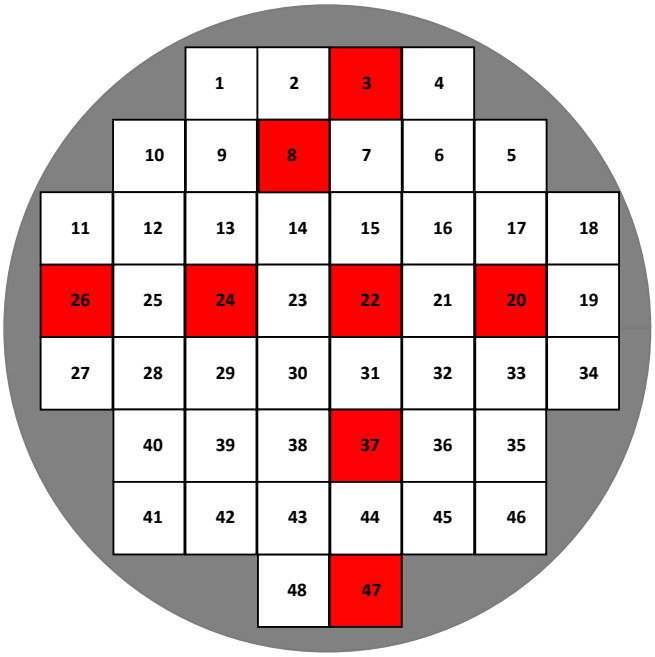


Figure 2. Wafermap illustrating all potentially viable dies; Dies marked red indicate the assessment of propagation loss and bend loss through the implemented test structures.

Propagation loss assessment utilized the cutback method across three waveguide structures with a length disparity of 1.2 cm. The Lambert-Beer law enabled the calculation of attenuation constants (α), crucial for data interpretation. Due to known fabrication variations, the assumption of consistent grating coupler efficiency in the NIR regime enhanced precision. Cross-verification using spirals of varying lengths provided contingency and facilitated accurate tree-based modelling.[6]

Bend losses, intertwined with propagation losses, were evaluated considering mode mismatch-induced losses. Compared to a reference 70 μm spiral, spirals with equal total lengths but varying bend radii enabled relative

measurements. The 90° bend loss (BL_x) for each waveguide width was determined using measurements from red dies in Figure 2. These dies' average propagation loss values and corresponding standard deviations were crucial for subsequent tree-based modelling.[6]

3. FABRICATION STRATEGY AND PROCESS

The fundamental layer stack comprises silicon nitride (Si_3N_4) waveguides within a silicon dioxide (SiO_2) undercladding and an uppercladding of the same material. Employing various techniques in a full-factorial Design of Experiments (DOE), the deposition steps were orchestrated for tree-based modelling. This systematic approach comprehensively explores the influence of multiple factors on desired outcomes, such as propagation and bend losses. The three layers, with two variations in each deposition, resulted in 40 investigated wafers, ensuring a minimum statistical sample size.[9]

The layers, thickness, refractive index (RI), and absorption (Abs) were characterized using an M-2000UI Woollam ellipsometer with a Cauchy model, incorporating Urbach absorption. Roughness measurements of the undercladding surface, waveguide top-surface, and sidewall were determined using the Anton Paar Tosca 400 Atomic Force Microscope. Different undercladding options were employed, including high-density plasma-enhanced chemical vapour deposition (HD-PECVD) SiO_2 and thermally grown SiO_2. The Si_3N_4 waveguide layer was deposited through plasma-enhanced chemical vapour deposition (PECVD) and low-pressure chemical vapour deposition (LPCVD). At the same time, the uppercladding was realized as HD-PECVD SiO_2 or sputtered SiO_2 using polycrystalline Si as the target material.[2,9]

Roughness data evaluation involved detrending using Gwyddion software and subsequent steps for waveguides. An initial model with an anisotropic, separable autocovariance kernel faced stability issues, including a 2D exponential kernel and a white noise kernel. For robustness, spectral analysis employed multitaper functions based on the discrete prolate spheroidal sequence (DPSS). This comprehensive approach addressed complexities in Atomic Force Microscope (AFM) data, particularly misalignment-related challenges, focusing on surface and sidewall roughness measurements.[2] Data is presented in Tables 1 and 2. For completeness, the layer stacks resulting from the 2x3 full-factorial DOE are summarized in Table 3.

Table 1. Selection of thin film layers for the Design of Experiments (DOE) on the silicon nitride waveguide platform. Measurements of thickness, refractive index, and absorption conducted using the M-2000UI Woollam ellipsometer.

	Thickness	RI @ 850 nm / 1	Abs @ 850 nm / 1
Thermal SiO_2 (undercladding)	2.2 μm ± 120 nm	1.50 ± 0.05	~ 0
HD-PECVD SiO_2 (undercladding after CMP)	2.2 μm ± 220 nm	1.46 ± 0.05	~ 0
HD-PECVD SiO_2 (uppercladding)	2.4 μm ± 150 nm	1.46 ± 0.05	~ 0
Sputtered SiO_2(uppercladding)	2.4 μm ± 120 nm	1.48 ± 0.05	~ 0
PECVD Si_3N_4 (Waveguide)	250 nm ± 15 nm	1.91 ± 0.05	$\sim 3 \times 10^{-6}$
LPCVD Si_3N_4 (Waveguide)	250 nm ± 15 nm	1.98 ± 0.05	$\sim 1 \times 10^{-6}$

Table 2. Thin film layer selection employed in the Design of Experiments (DOE) for the silicon nitride waveguide platform. Evaluation of root mean square (RMS) roughness, undercladding correlation length (CL), waveguide sidewall correlation length along the waveguide direction (CLZ), and waveguide sidewall correlation length perpendicular to the waveguide direction (CLX) conducted using the Anton Paar TOSCA 400 Atomic Force Microscope.

	RMS / nm	CLZ / nm	CLX / nm	CL / nm
Thermal SiO_2 (Undercladding)	0.28	-	-	10.5
HD-PECVD SiO_2 (Undercladding; after CMP)	0.64	-	-	7.1
PECVD sidewall Si_3N_4 (Waveguide)	2.91	126.10	819.96	-
PECVD top-surface Si_3N_4 (Waveguide)	1.98	17.14	29.29	-
LPCVD sidewall Si_3N_4 (Waveguide)	2.04	141.99	1390.52	-
LPCVD top-surface Si_3N_4 (Waveguide)	0.87	104.83	417.37	-

Table 3. Layer stacks used in the 2x3 full-factorial DOE

	Undercladding	Waveguide layer	Uppercladding
Stack 1	Thermal SiO_2	LPCVD Si_3N_4	HD-PECVD SiO_2
Stack 2	Thermal SiO_2	LPCVD Si_3N_4	Sputtered SiO_2
Stack 3	Thermal SiO_2	PECVD Si_3N_4	HD-PECVD SiO_2
Stack 4	Thermal SiO_2	PECVD Si_3N_4	Sputtered SiO_2
Stack 5	HD-PECVD SiO_2	LPCVD Si_3N_4	HD-PECVD SiO_2
Stack 6	HD-PECVD SiO_2	LPCVD Si_3N_4	Sputtered SiO_2
Stack 7	HD-PECVD SiO_2	PECVD Si_3N_4	HD-PECVD SiO_2
Stack 8	HD-PECVD SiO_2	PECVD Si_3N_4	Sputtered SiO_2

The detailed fabrication process is illustrated in Figure 3. The process utilized a silicon wafer as the substrate material, with a thickness of $725 \pm 15\ \mu m$. Before deposition, meticulous cleaning using wet chemistry methods eliminated undesired particle contamination. Subsequent steps involved depositing either a 2.4 μm-thick layer of SiO_2 using HD-PECVD or achieving a 2.2 μm-thick SiO_2 through wet chemical oxidation. Chemical-mechanical polishing enhanced uniformity for HD-PECVD SiO_2, followed by the deposition of a 250 nm thick Si_3N_4 waveguide layer, either via LPCVD or PECVD. Ellipsometer measurements monitored thickness and stress after each deposition.

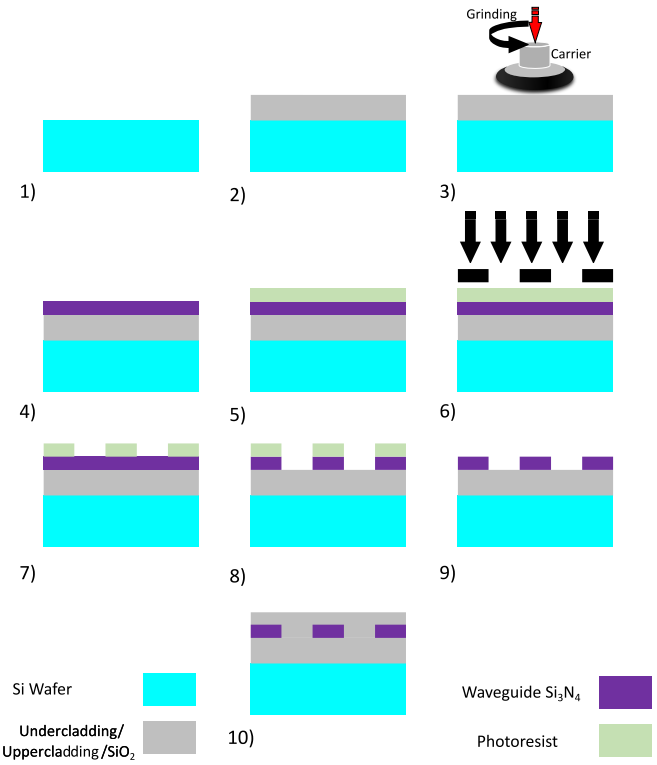


Figure 3. Fabrication Process Flow: 1) a pristine Si wafer as the substrate; 2) Deposition of SiO_2 as the undercladding; 3) Implementation of CMP on the undercladding for surface smoothing (conditional on the deposition of HD-PECVD SiO_2); 4) Deposition of Si_3N_4 as the waveguide layer; 5) Spin coating of Photoresist; 6) UV Exposure; 7) Photoresist development; 8) Precision structuring of Si_3N_4 waveguides through reactive ion etching; 9) Strip-off photoresist; 10) Deposition of SiO_2 as the uppercladding via sputtering or HD-PECVD.

Advanced deep ultraviolet (DUV) lithography and a two-step reactive ion etching process were employed during the structuring phase. Post-etch SEM analysis ensured the removal of resist residuals and confirmed critical dimensions. A 2.4 μm-thick uppercladding layer was sputtered, which is beneficial for sensing applications. Alternatively, HD-PECVD SiO_2 with a thickness of 2.4 μm was utilized. The final steps involved ellipsometer assessments of the uppercladding's thickness and supplementary wafer stress evaluations, laying the groundwork for subsequent measurements.[6,10]

4. EVALUATION OF DOE UTILIZING RANDOM FORESTS

Random Forest regression, the ensemble technique based on CART, enhances predictive accuracy by aggregating multiple regression trees, each built from a bootstrap sample of the dataset. It introduces additional randomness by selecting a random subset of predictors for tree splits, effectively reducing the variance of single-tree models without notably increasing bias. This method benefits from out-of-bag (OOB) samples—data not included in the bootstrap sample—to estimate prediction error internally, providing an unbiased error estimation without the need for a separate validation set (Breiman, 2001).[11]

The strength of Random Forest lies in its robustness against overfitting, attributed to the law of large numbers as the ensemble approach converges to a stable solution. Despite its effectiveness in improving prediction accuracy across diverse datasets, the model's complexity renders it a "black box," challenging interpretability and variable importance visualization. Efforts to develop methodologies for extracting insights from Random Forest have partially mitigated these challenges, yet the interpretability issue remains.[7] In essence, Random Forest stands out for its capacity to combine the simplicity of decision trees with the power of ensemble learning, making it a valuable tool for regression tasks. Its self-assessment through OOB samples and resistance to overfitting underscore its utility in predictive modelling, despite interpretability hurdles.[11]

In the exploration of data mining processes, an assessment was conducted focusing on the mean propagation loss across 40 distinct wafers as part of a Design of Experiments (DOE). Each wafer was characterized by six waveguide width parameters. This comprehensive analysis was facilitated through the use of the statistical computing environment R (Version 4.3.2), utilizing the randomForest package (Version 4.7.1.1), renowned for its efficacy in ensemble tree methodologies, across a dataset comprising 240 observations. The initial phase involved the construction of a regression tree, positioning the mean propagation loss per wafer as the target variable, and employing categorical predictors such as layer information and waveguide width.[12]

To enhance the model's accuracy in leveraging Random Forests for data analysis, optimizing the number of trees (ntree) and the number of features considered at each split (mtry) is crucial for balancing model accuracy with computational efficiency. The optimal ntree ensures adequate model performance without incurring unnecessary computational costs, identified through cross-validation where model accuracy is plotted against the number of trees to find a plateau point.

Similarly, the optimal mtry value, representing the number of features considered for each split, is efficiently executed using the tuneRF function from the randomForest package. This function iteratively evaluates different mtry values by doubling them starting from a lower bound, aiming to minimize the out-of-bag (OOB) error rate until the improvement falls below a predefined threshold (default: 5%). The tuneRF function's utility is demonstrated in its ability to automatically adjust mtry, thereby enhancing model performance through a systematic search process. This methodology facilitates the identification of an optimal mtry value, directly contributing to the refinement of the Random Forest model by balancing accuracy and generalization capability. The resultant mtry parameter, indicative of the model's reduced error rate, is subsequently employed in the final model training phase, reinforcing the model's predictive robustness in analytical applications.[12]

4.1 RANDOM FOREST BASED EVALUATION OF PROPAGATION LOSSES OF THE LAYER-DOE WAFER

In this section towards the evaluation of propagation losses across the Layer-DOE wafers, a systematic analysis employing the Random Forest algorithm was conducted to ascertain the statistical relationship between mean propagation losses and a set of explanatory variables, including waveguide dimensions and cladding material compositions. The analysis commenced with the processing of data within the R statistical computing environment, utilizing the randomForest package to instantiate the predictive model. The response variable, mean propagation loss over the wafer , while the predictors comprised physical attributes such as Width and the layer material undercladding, waveguide, and uppercladding. A random forest model was created based on the whole dataset, by deploying 1000 decision trees with an mtry parameter of 4, indicating the subset of variables considered for splitting at each node. This model was subject to empirical evaluation, with its performance gauged through the analysis of the decrease and graphical interpretation of the OOB mean square error (MSE) over tree complexity. The optimization of the mtry paramter was done via the tuneRF function was employed to systematically adjust the mtry parameter, aiming to refine the model by minimizing the out of bag (OOB) error rate.

A visualization of ensemble methods is in general quite difficult, since the aggregating of the trees is not quite visually adaptable. A visualization is to contrast the observed propagation losses with the predicted out-of-bag (OOB) propagation losses, highlighting the model's varying accuracy in predicting losses across different magnitudes as shown in Figure 4.

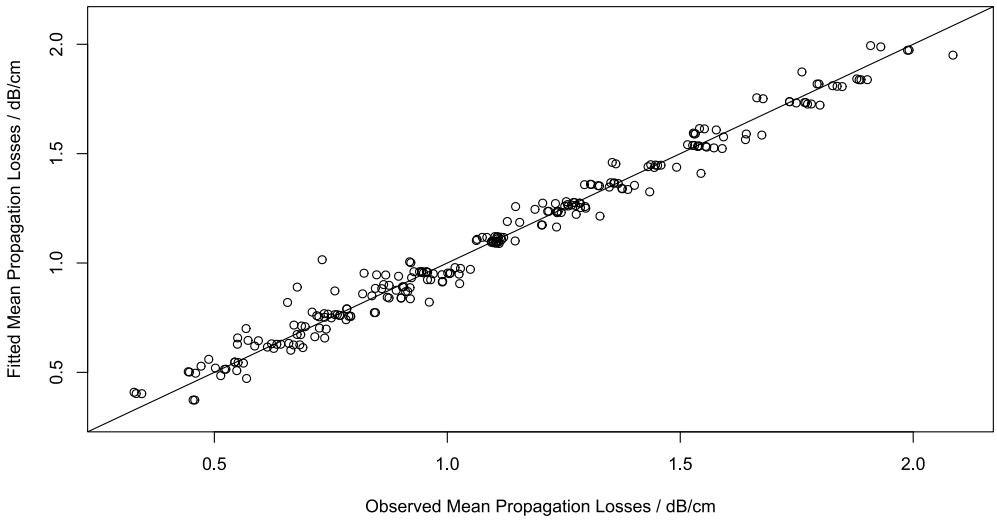


Figure 4. Comparative Visualization of Observed Versus Predicted Propagation Losses by Random Forest Regression Analysis of Propagation Losses Across Full-Factorial DOE Layers referencing the design in Table 3.

The qualitative assessment in Figure 4 shows a very good fit. In assessing the performance of our Random Forest model for predicting propagation losses, two quantitative key metrics were utilized: the Root Mean Squared Error (RMSE) and the pseudo R^2 value. The RMSE, a widely recognized measure for quantifying the accuracy of predicted values, was computed as 0.06 dB/cm. This low RMSE indicates a high level of precision in the model's predictions relative to the actual observed losses, underscoring the model's efficacy in capturing the underlying patterns of propagation loss across the dataset. In a previous CART study, an RMSE of0.1303 dB/cm was reached in the model where all possible predictors where included.[6]

Furthermore, to evaluate the proportion of variance in the observed propagation losses explained by the model, a pseudo R^2 value was calculated. The total sum of squares (TSS), representing the total variance in the observed data, and the residual sum of squares (RSS), representing the variance unexplained by the model, were determined. The pseudo R^2 value was subsequently derived as $1 - \frac{RSS}{TSS}$, yielding a value of 0.981. This value suggests that the Random Forest model accounts for approximately 98% of the variance in propagation losses, indicating an exceptional fit to the data. In the preceding CART study, an apparent R^2 value of 0.8912 was obtained under this model conditions however, here, it is important that the comparability between different types of R^2 is not always straightforward.[6,13]

To go into the significance of various predictors in estimating propagation losses, variable importance plots of the random forest model were created and are visible in Figure 5.

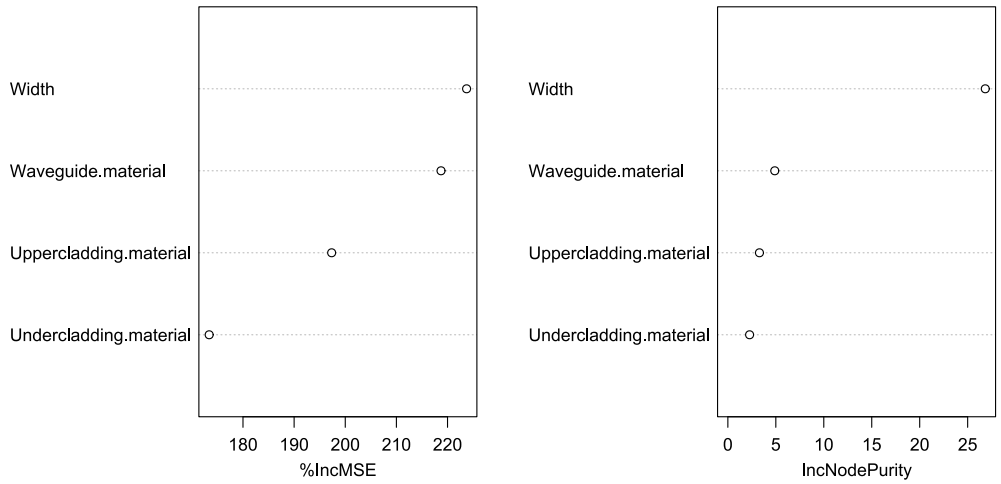


Figure 5. Impact of Predictors on Propagation Loss Estimation Accuracy. This visualization ranks predictors based on their influence on Random Forest model accuracy, using %IncMSE and %IncNodePurity metrics.

The %IncMSE metric in Random Forest models signifies the increase in Mean Squared Error (MSE) resulting from the permutation of a predictor's values, offering a gauge of the variable's impact on prediction accuracy. Permuting a predictor underlines the model's dependency on it for precision, with a notable rise in MSE underscoring the variable's criticality. This metric serves as a direct indicator of importance, with higher values spotlighting essential predictors and emphasizing the significance of data quality and feature engineering for model efficacy.

Conversely, %IncNodePurity quantifies a predictor's role in enhancing the homogeneity, or purity, of decision tree nodes, reflecting its capacity to generate clear, homogenous groups that bolster prediction accuracy. It accumulates the gains in purity across splits on the predictor, weighted by the sample size affected, highlighting variables that markedly refine decision-making within the model. High %IncNodePurity values identify predictors crucial for data segmentation, underpinning their value in model optimization. Together, %IncMSE and %IncNodePurity provide critical insights for feature selection, underlining the intertwined roles of individual predictors in model accuracy and decision clarity. In the following paragraph, thorough interpretation of these metrics for each variable is listed according to Figure 5 :

- **Width**
  - **%IncMSE: 223.7269** - Indicates that the width of the waveguide is crucial for predicting propagation loss, with a substantial increase in Mean Squared Error when its value is permuted. This highlights its significant impact on the model's predictive accuracy.
  - **IncNodePurity: 26.836240** - Shows that width plays a significant role in creating more homogeneous subsets, being a powerful predictor in the model by contributing to the decrease in model impurity.
- **Undercladding Material**
  - **%IncMSE: 173.3892** - Suggests a notable impact on the model's predictive accuracy, though lesser than width, indicating its importance in predicting propagation losses.
  - **IncNodePurity: 2.265338** - Indicates a lesser contribution to the overall homogeneity of the model's predictions compared to width.
- **Waveguide Material**

  - **%IncMSE: 218.7266** - Similar to width, shows a significant effect on increasing MSE, underlining the critical role of waveguide material in the accurate prediction of propagation losses.
  - **IncNodePurity: 4.889715** - With a moderate increase in node purity, it contributes to the model's splitting decisions, affecting the homogeneity of the resulting subsets.

- **Uppercladding Material**
  - **%IncMSE: 197.3397** - Indicates its impactful role on the model's MSE, thus highlighting its relevance in the accurate prediction of propagation losses.
  - **IncNodePurity: 3.292663** - Demonstrates its contribution to increasing the homogeneity of model predictions, albeit less so than the width but significant nonetheless.

The interpretation of the variable importance is now as follows, also with respect to the %IncMSE value which is a more common metric in regression tasks: The model demonstrates that waveguide width is the most critical factor, with an and a %IncMSE of 223.7269 when predicting propagation loss. This significant increase in MSE upon permutation and its considerable contribution to model homogeneity (IncNodePurity: 26.836240) emphasize width's paramount importance in PIC design. The %IncMSE for width corroborates the physical principle that the waveguide's geometric dimensions, are crucial for predicting the optical losses thoroughly due to the high influence of the width on the confinement, which determines the propagation loss quite heavily. For example, in case of high confinement which corresponds to higher widths, the waveguide material absorption dominates. Whereas, decreasing the width, keeping everything else constant, the cladding as well as the interface properties between the cladding and the waveguide, like the roughness on the topsurface and the sidewall, dominate.[14] In terms of the CART study in[6] this domination of the width is qualitively also given.

With an %IncMSE of 218.7266, the waveguide material closely follows width in its importance for accurate loss prediction. This underscores the material's role in affecting the aforementioned internal absorption and scattering losses, necessitating careful selection to predict and optimize PIC performance.

Although less influential than width and waveguide material, the cladding materials still significantly impact propagation losses (%IncMSE of 173.3892 for undercladding and 197.3397 for uppercladding). Their contributions to model homogeneity, as indicated by IncNodePurity scores, highlight their roles in the overall PIC structure, affecting loss characteristics to varying degrees. Here it is important to mention that the uppercladding has a bigger influence due to fact that either sputtered oxide or HDP oxide is used whereas sputtered oxide is leading to higher losses, since there is a quite enourmous difference in the conformality of the two uppercladding oxides.[15] The variable importances of the materials is also in quite good accordance with the CART results.[6]

4.2 RANDOM FOREST BASED EVALUATION OF BEND EXCESS LOSSES OF THE LAYER-DOE WAFER

In this section, the evaluation of bend excess losses across Layer-DOE wafers through a systematic analysis using the random forest algorithm is carried out. Data processing commenced again within the R statistical environment, leveraging the `randomForest`.

The response variable under investigation is the mean bend excess loss over the wafer, while predictors include physical attributes such as Width and radius and the layer materials (undercladding, waveguide, and uppercladding). A Random Forest model, incorporating 1000 decision trees with an `mtry` parameter of 5, was trained on the entire dataset. This parameter indicates the subset of variables considered for splitting at each node. Model performance was empirically evaluated by analysing the decrease in out-of-bag (OOB) mean squared error (MSE) as a function of tree complexity, and the `mtry` parameter was optimized using the `tuneRF` function to minimize the OOB error rate.

In Figure 6 the observed bend excess losses with predicted OOB bend excess losses are highlighting the model's prediction accuracy across different loss magnitudes are displayed.

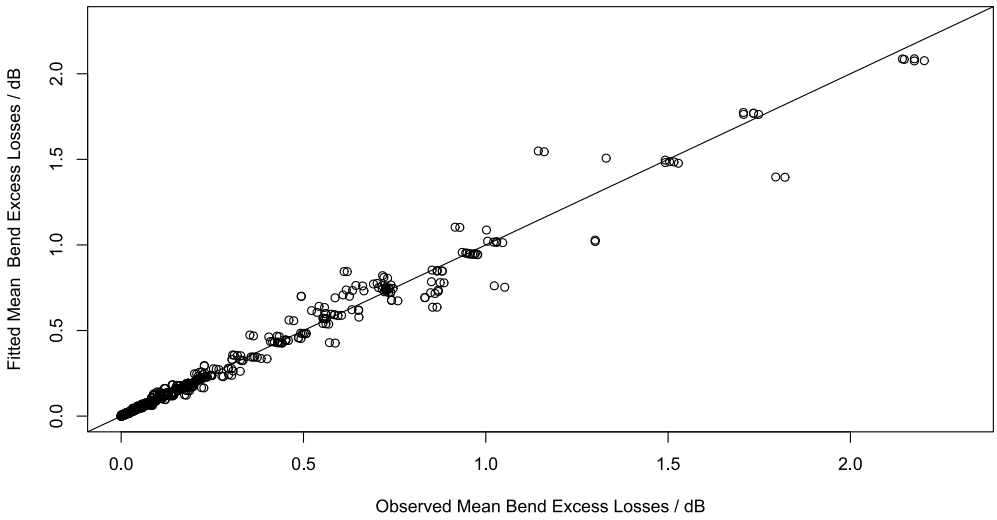


Figure 6. Comparative Visualization of Observed Versus Predicted Bend Excess Losses by Random Forest Regression Analysis of Propagation Losses Across Full-Factorial DOE Layers Referencing the Design in Table 3.

The qualitative assessment in Figure 6 demonstrates an excellent fit between observed and predicted losses. Quantitatively, the Random Forest model's performance was evaluated using again the Root Mean Squared Error (RMSE) and the pseudo R^2 value. An RMSE of 0.04 dB signifies a high precision level in the model's predictions but since the minimal values are on the order of magnitude of 10^-5, this accuracy is not high enough. Although the pseudo R^2 value, calculated as $1 - \frac{RSS}{TSS}$, was found to be 0.982, probably modelling different radius parameters is necessary.

Variable importance plots for the Random Forest model were generated to examine the significance of different predictors in estimating bend excess losses, as shown in Figure 7. These plots rank predictors based on their influence on model accuracy through %IncMSE and %IncNodePurity metrics, providing insights into feature selection and model refinement.

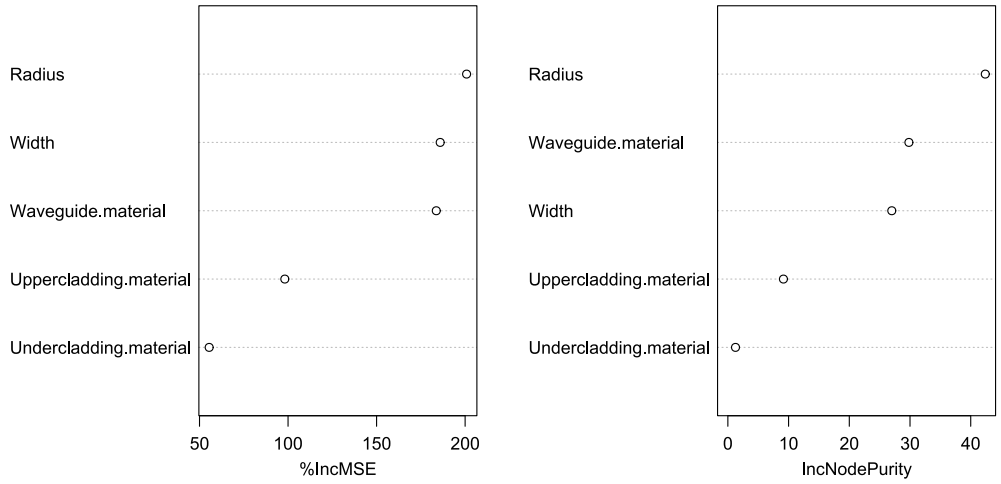


Figure 7. Impact of Predictors on Bend Excess Loss Estimation Accuracy. This visualization ranks predictors based on their influence on Random Forest model accuracy, using %IncMSE and %IncNodePurity metrics.

Given the model's performance on predicting bend excess loss, we observe the following impacts of various parameters:

- **Radius**
  - **%IncMSE: 200.82802** - Indicates that the radius of the waveguide bend has a profound impact on the predictive accuracy for bend excess loss, with a significant increase in Mean Squared Error when its value is altered, underscoring its crucial role in the model.
  - **IncNodePurity: 42.395430** - Shows that radius is a powerful predictor in the model by contributing to a substantial increase in the homogeneity of the subsets, highlighting its dominant impact on model impurity reduction.
- **Width**
  - **%IncMSE: 185.92242** - Suggests a notable impact on the model's predictive accuracy for bend excess loss, indicating its importance, though slightly lesser than radius, in predicting bend excess losses.
  - **IncNodePurity: 26.997370** - Indicates that width plays a significant role in enhancing the model's homogeneity, albeit not as dominant as radius.
- **Undercladding Material**
  - **%IncMSE: 55.45619** - Shows a lesser impact on the model's predictive accuracy for bend excess loss compared to radius and width, but still underlines the importance of undercladding material in accurate predictions.
  - **IncNodePurity: 1.256393** - Indicates a minimal contribution to the overall homogeneity of the model's predictions, suggesting its lesser significance in the model compared to other parameters.
- **Waveguide Material**
  - **%IncMSE: 183.70948** - Similar to width, shows a significant effect on increasing MSE for bend excess loss, highlighting the critical role of waveguide material in accurate predictions.
  - **IncNodePurity: 29.818211** - With a significant increase in node purity, it underscores the importance of waveguide material in the model's decision-making processes.
- **Uppercladding Material**
  - **%IncMSE: 98.16804** - Indicates its impactful role on the model's MSE for bend excess loss, thus underlining its relevance in accurate prediction, albeit less so than radius, width, and waveguide material.
  - **IncNodePurity: 9.152036** - Demonstrates its contribution to increasing the homogeneity of model predictions to a significant extent, but less impactful than radius and waveguide material.

The detailed analysis of factors influencing bend excess loss in photonic integrated circuits reveals, anologous to the propagation loss analysis, the crucial roles of both geometric parameters—specifically, the radius and width of waveguides—and the material properties of the waveguide and cladding layers. The focus is again put on the %IncMSE value for the variable importance since this is used for regression cases when comparing it to CART variable importance quantitatively. The bend radius emerges as the most significant factor affecting predictive accuracy for bend excess loss, with its alteration leading to a substantial increase in Mean Squared Error %IncMSE: 200.82802), highlighting its pivotal role in the model , which is also due to the influence on the bend loss over multiple orders of magnitude. This is closely followed by waveguide width, which also significantly impacts the model's predictive accuracy (%IncMSE: 185.92242), though to a slightly lesser extent than the radius. These findings underline once again the importance of optimizing geometric dimensions to predict the waveguide performance properly and minimize losses effectively. The intricate relationship between

waveguide width, confinement, bend radius, and mode mismatch plays a crucial role in the optimization of photonic integrated circuits (PICs), revealing a complex interplay that significantly impacts bend excess loss. The essence of this complexity lies in how these factors interact to influence light propagation and loss mechanisms within the waveguide structure. Waveguide width and bend radius are primary geometric parameters that dictate the confinement of light within the waveguide. The width of the waveguide directly affects the confinement, with narrower waveguides generally leading to tighter confinement of the optical mode. This tight confinement is essential for efficient light propagation, particularly in bends, where avoiding excessive loss is crucial. However, as the confinement increases with decreasing width, it also heightens the sensitivity of the waveguide to imperfections and introduces challenges in managing mode overlap and mismatch, especially in bends. The bend radius introduces another layer of complexity to this scenario. Smaller radii can exacerbate the challenges posed by tight confinement, as the curvature of the bend can lead to a significant mode-overlap mismatch. This mismatch occurs because the optical mode's natural propagation path is distorted by the bend, leading to inefficiencies and increased loss. In terms of importance, the radius can be considered a factor of even higher significance compared to width in certain contexts. This is primarily because the mode-overlap mismatch induced by the bend radius can significantly impact the optical mode's integrity, leading to pronounced losses that are particularly challenging to mitigate. The dependencies between these factors are intricate: while waveguide width determines the baseline confinement of the optical mode, the bend radius can dramatically influence how this confinement translates into efficient or lossy light propagation in curved sections of the PIC. The mode mismatch, as a consequence of these geometric parameters, underscores the delicate balance required in designing waveguides that can efficiently navigate the trade-offs between confinement, bend radius, and minimal mode distortion.[2,6]

In terms of material properties, the waveguide material stands out for its notable effect on increasing Mean Squared Error (%IncMSE: 183.70948), nearly mirroring the impact of waveguide width and indicating its critical role in accurate loss prediction. The importance of cladding materials, though less pronounced than that of radius or waveguide material, is still significant, with undercladding and uppercladding materials affecting the model's accuracy (%IncMSE: 55.45619 for undercladding and 98.16804 for uppercladding), albeit to varying degrees. This underscores the need for careful selection of material properties, considering how these interact with the waveguide geometry to influence bend loss characteristics. The reasons for the loss determining factors are again the waveguide absorption and roughness which play an importnant role and the confromality of the uppercladding which is different and can act when non-conformal as scattering center.

Upon analysing Figure 6, it becomes evident that the data exhibits significant imbalance, suggesting a need for stratification based on different bend radii. Given that bend radius emerges as the dominant predictor, with bend excess loss varying across several orders of magnitude, partitioning the data by radii appears to be a prudent approach at this juncture. This strategy will facilitate a more nuanced understanding of the influence of bend radius on excess loss, thereby enhancing the accuracy and relevance of the predictive models.

5. CONCLUSION AND OUTLOOK

In the comprehensive investigation of propagation and bend excess losses within Layer-DOE wafers, an advanced statistical analysis employing the Random Forest algorithm was meticulously conducted. This study focused on unravelling the complex statistical relationship between optical losses and an array of explanatory variables, including waveguide dimensions, cladding material compositions, and notably, the operational wavelength of 850 nm. Utilizing the `R` statistical computing environment, the analysis leveraged the `randomForest` package to construct a predictive model, incorporating 1000 decision trees with an `mtry` parameter of 4. This approach facilitated a nuanced empirical evaluation, with the OOB MSE serving as a key metric for assessing model accuracy.

The quantitative findings of this investigation are noteworthy. The RMSE for propagation losses was determined to be an impressively low 0.06 dB/nm, indicating the high precision of the model's predictions in capturing the actual observed losses. This level of precision is particularly significant given the operational wavelength of 850 nm, where the dynamics of light-matter interaction are critically relevant. Furthermore, the model exhibited remarkable explanatory power, as evidenced by a pseudo R^2 value of 0.981, suggesting that it accounts for approximately 98% of the variance in propagation losses.

The analysis also delved into the impact of various predictors on model accuracy through variable importance plots. The waveguide width emerged as a critical predictor, with a %IncMSE value of 223.7269, highlighting its significant role in determining optical losses. This underscores the importance of geometric dimensions, especially at the operational wavelength of 850 nm, aligning with the physical principles that emphasize the influence of waveguide width on light confinement and propagation loss.

For bend excess losses is indicated by an RMSE of 0.04 dB/nm, which is not considered sufficiently accurate since the minimal values are on the order of 10^{-5}. Despite this, the model demonstrates considerable explanatory power with a pseudo R^2 value of 0.982. Moreover, with an adjusted `mtry` of 5, the model's analysis highlighted the critical importance of the waveguide bend radius in accurately predicting bend excess losses—evidenced by a %IncMSE value of 200.82802—especially pertinent due to the effects of curvature on light propagation in photonic circuits at the given wavelength.

The insights from this analysis illuminate the intricate interdependencies between waveguide characteristics and optical performance at the critical wavelength of 850 nm. Moreover, they underscore the effectiveness of the Random Forest algorithm in providing a deep understanding of these dynamics. The derived quantitative metrics, such as RMSE and pseudo R^2, alongside the thorough evaluation of variable importance, lay a solid foundation for future endeavours. These include optimizing waveguide design and material selection to minimize losses and enhance the performance of photonic devices, particularly for applications operating around the pivotal wavelength of 850 nm. Through such sophisticated statistical analyses, this study paves the way for significant advancements in the fabrication and optimization of photonic integrated circuits, opening new avenues for enhancing device efficiency and functionality in technological applications.

Moving forward, the rigour of analysing optical losses in photonic waveguides can be enhanced by segmenting the data based on the main predictors: waveguide width for propagation loss and bend radius for bend excess loss. This focused approach allows for a more nuanced understanding of how these factors independently affect loss mechanisms. Moreover, adopting a holistic data analysis strategy, including train-test splits for model validation, hyperparameter optimization for refining model performance, and the selection of appropriate evaluation metrics, will ensure comprehensive model assessment. Implementing ensemble methods and establishing baselines for comparison will further bolster the analytical framework, offering a pathway to more accurately predict and mitigate optical losses. Such systematic and granular analysis promises to advance the development of photonic integrated circuits, optimizing their efficiency and functionality.

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